A Digital Compute-in-Memory Macro Featuring Two's Complement Multiplication for LSTM-based Biomedical Signal Classification

Jinhai Hu*†, Wang Ling Goh†, Yuan Gao*
*Institute of Microelectronics (IME), Agency for Science, Technology and Research (A*STAR), 2 Fusionopolis Way, Innovis #08-02, Singapore 138634, Republic of Singapore
†School of Electrical and Electronic Engineering, Nanyang Technological University (NTU), Republic of Singapore
Email: hujh@ime.a-star.edu.sg, ewlgoh@ntu.edu.sg, gaoy@ime.a-star.edu.sg

2025 IEEE International Symposium on Circuits and Systems (ISCAS) | 979-8-3503-5683-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ISCAS56072.2025.11044146

***Abstract*—This paper presents a digital compute-in-memory (DCIM) macro that supports two's complement multiplication, specifically designed for processing electrocardiogram (ECG) signals using a Long Short-Term Memory (LSTM) neural network. Two distinct bitcell computing mechanisms are introduced: one for two's complement bit-serial recurrent inputs using a 6T SRAM bitcell with two transmission gates (TGs) for outputting a weight bit or its complement, and another for encoded one-hot ECG inputs using an 8T bitcell to output weight values based on "1" detection in the input. Each column of bitcells performs multiply-and-accumulate operations, computing bitwise vector-matrix multiplication between inputs and SRAM-stored weights. Partial sums generated by columns of DCIM cells are processed through an adder tree controlled by a shift register, yielding the final LSTM gate-sum result via a parallel adder. The proposed DCIM macro enhances hardware efficiency by reducing transistor count and supports precise two's complement multiplication. It achieves 96.9% accuracy on a 5-class classification task, using 32-level one-hot ECG input and an INT5 quantized LSTM neural network.**

Index Terms*— *Digital compute-in-memory (DCIM), SRAM, ECG classification, Long Short-Term Memory (LSTM)

I. Introduction

Artificial intelligence, data-driven computing, machine learning, and optimization tasks have driven significant interest in developing hardware accelerators specifically designed to process artificial neural networks using massively parallel data. These accelerators shift the burden of vast data processing from the cloud to the edge, addressing increasing concerns around communication bandwidth, latency, privacy, and security. Compute-in-Memory (CIM) stands as a pivotal approach in this context, offering a pathway to enhanced energy efficiency and reduced data transfer latency, which are essential for edge computing.

Analog CIM, commonly implemented on resistive memory [1], [2], provides benefits such as reduced area and power consumption, making it attractive for low-power applications. However, it suffers from precision losses due to the non-ideal characteristics of analog components, which can limit its effectiveness in applications that demand high accuracy. In contrast, Digital CIM (DCIM), particularly on SRAM, preserves full precision and is capable of delivering accurate computations. DCIM leverages pure digital logic, ensuring scalability and robustness against process, voltage, and temperature (PVT) variations [3] – [6], making it more suitable for precision-critical applications.

This work was supported by the Agency for Science, Technology and Research (A*STAR), Singapore under the High Linearity Silicon Germanium Photonic Modulator for 6G Analog Radio over Fiber Project, Grant No. M24M8b0004.

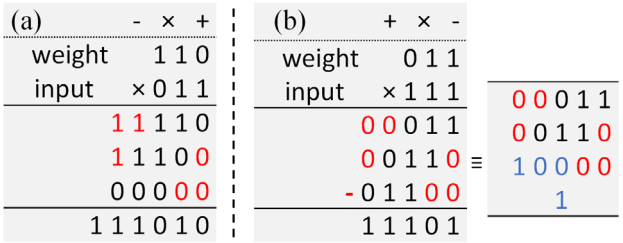


Fig. 1. Examples of 3-bit two's complement multiplication, from which (a) is positive input multiply with negative weight, while (b) is the counterpart.

One such application is ECG classification, essential for the early detection and monitoring of cardiovascular diseases (CVDs). Conventional solutions like Holter monitors present challenges due to discomfort, interference, and the need for data downloads at medical facilities, underscoring the need for real-time, accurate ECG monitoring and classification [7] – [9]. Thus, the precise and reliable computations enabled by DCIM make it a compelling choice for next-generation ECG monitoring devices aimed at improving patient care and diagnostic accuracy.

This paper introduces a DCIM design optimized for ECG signal processing, which requires precise and reliable computations. Traditional DCIM approaches face challenges, such as the need for an adder tree for each column of weights and the complexity of implementing two's complement multiplication, which typically demands numerous gates. To address these issues, our design reuses a single adder tree, trading off clock cycles due to the relatively low ECG signal frequency. This unconventional computing mechanism further simplifies the hardware by employing transmission gates (TG) to control the on/off states of SRAM cells representing weights, effectively bypassing the need for direct multiplication. Through this mechanism, it achieves the same process of Baugh-Wooley multiplication algorithm [10], as shown in Fig. 1. It uses bit inversion and sign extension to simplify the handling of negative values, making it well-suited for hardware implementations of two's complement arithmetic. This design demonstrates that DCIM is a compelling alternative to analog CIM for applications where both accuracy and efficiency are critical.

The paper is organized as follows: Section II introduces the proposed DCIM macro for LSTM-based neural network. The results of the proposed DCIM macro for ECG signal classification and benchmarking are shown in Section III, followed by a summary in Section IV.

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on September 29,2026 at 12:18:01 UTC from IEEE Xplore. Restrictions apply.

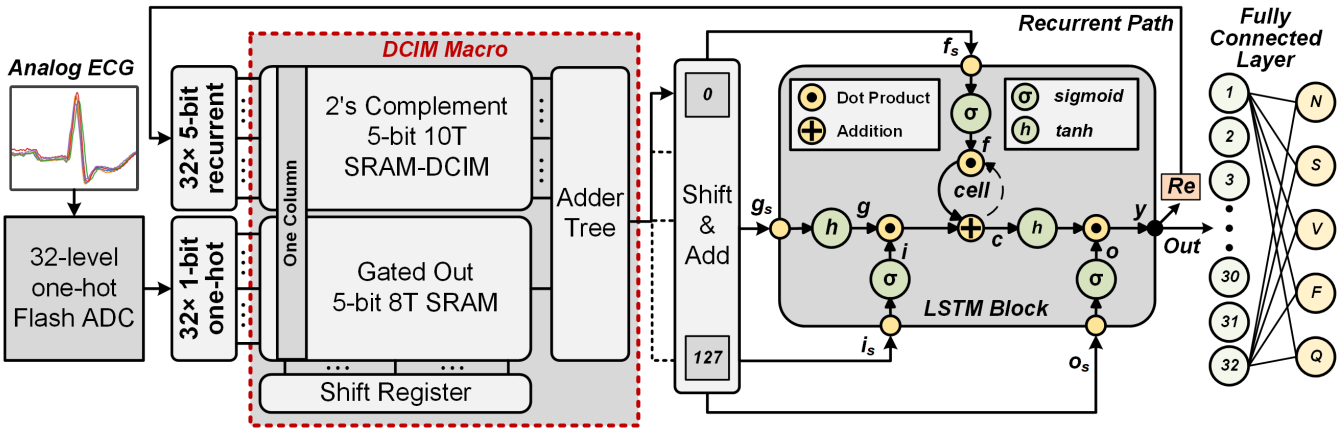


Fig. 2. Proposed ECG classification framework through 2's complement DCIM macro and LSTM neural network.

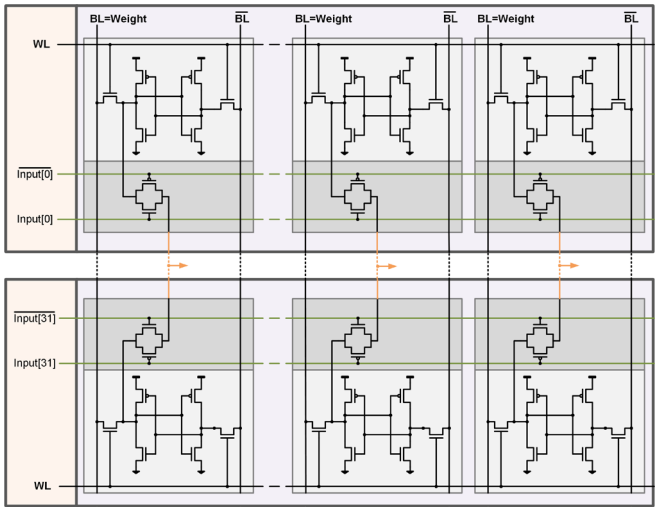


Fig. 3. Gated-out SRAM architecture for encoded multi-level one-hot ECG input signal.

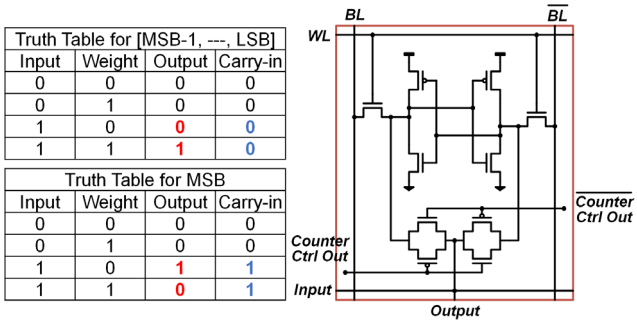


Fig. 4. (left) Truth table for two's complement multiplication. (right) 10T SRAM-DCIM bitcell for recurrent input multiplication.

II. Proposed Models

The block diagram of the proposed ECG classification framework is shown in Fig. 2. It comprises four major function blocks: a multi-level one-hot flash analog-to-digital converter (ADC), a DCIM macro with two distinct bitcell computing mechanisms, an LSTM block, and a fully connected (FC) layer. Firstly, the ECG data is encoded into a 32-level one-hot array using an encoder-free ADC [11]. Then, the DCIM macro performs the vector-matrix multiplication for the four gate-sums in the LSTM layer. The LSTM block learns the ECG signal features in a time-series format, encoding them into a high-dimensional embedding vector. Finally, the FC layer was employed for category classification.

A. Multi-level One-hot Encoding

The dataset used in this work is the public available MIT-BIH dataset [12]. A multi-level one-hot encoding strategy is introduced in [11] to encode every numeric value in the single-lead ECG signal into a one-hot binary array with 2^N levels, where N is the quantization bit-width. The size of the encoded array at each time step is $2^N \times 1$, and only one bit is set to 1 while all others are set to 0. This encoding scheme brings several compelling benefits. Firstly, it simplifies the input data into a binary value vector with high sparsity. Secondly, it ensures that the number of Multiply-Accumulate (MAC) operations in matrix multiplication remains consistent comparing to taking conventional single-lead ECG as input. Lastly, this approach extracted more feature information from the input signal, ultimately yielding higher classification accuracy compared to the conventional methods.

In this work, N is set to 5 to convert the normalized ECG signal into 32 levels. Hence, a 32 × 1 one-hot array is generated at each time step as the input to LSTM layer.

B. DCIM Computation Mechanisms

The gate operations of the LSTM cell are illustrated in [13], represented by four distinct equations. In the hardware approach, these four operations can be consolidated into a single matrix multiplication shown in (1) to compute the four gate-sums. The weight dimension and input dimension are $(4 \times n, n + m)$ and $(n + m, 1)$, where n denotes the recurrent dimension, and m represents the input size. In this work, we have set both n and m to 32, representing the dimensions of the recurrent input (h_{t-1}) and ECG input (x_t), respectively.

$$\begin{bmatrix} f_s \\ g_s \\ i_s \\ o_s \end{bmatrix} = \begin{bmatrix} W_f & R_f \\ W_g & R_g \\ W_i & R_i \\ W_o & R_o \end{bmatrix} \times \begin{bmatrix} x_t \\ h_{t-1} \end{bmatrix} \quad (1)$$

To implement this matrix multiplication on DCIM, we employ two types of bitcell for the computation: one handles unsigned single-bit 32-level one-hot ECG input, while the other manages 32 signed INT5 recurrent input.

The gated-out SRAM architecture shown in Fig. 3 is designed to handle encoded multi-level one-hot ECG input signals efficiently. In this structure, each SRAM cell within a row is connected to a common word line (WL), which activates the cells for read or write operations. The bit lines (BL) are configured to store weights. Each bit from the input

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on September 29,2026 at 12:18:01 UTC from IEEE Xplore. Restrictions apply.

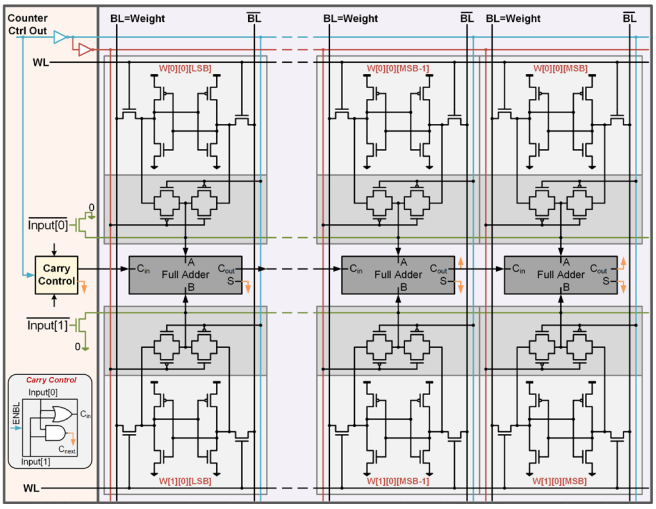


Fig. 5. DCIM architecture for bit-serial recurrent input multiplication.

vector (Input[0] to Input[31]) is mapped to a specific bit line, representing different levels of the signal. Since only one entry in the input vector is set to "1", the architecture can selectively gate the corresponding row of SRAM cells, enabling precise control over which cells participate in the data operation. This reduces unnecessary switching and potentially saves energy.

Based on the truth tables for two's complement multiplication shown in Fig. 4(left), the schematic of bitcell is shown in Fig. 4(right). The rules dictate that the weight values are stored statically in SRAM, while the input values are processed in bit-serial order, starting from the MSB down to the LSB. For all bits except the MSB, the multiplication process utilizes an "AND" gate with the carry-in set to zero. However, for the MSB, the output is the complemented value of the weight if both the input is set to one and the carry-in is set to one, referring to Fig. 1(b). The DCIM architecture for recurrent input multiplication is shown in Fig. 5. In this design, a control signal, "Counter Ctrl Out", is activated (set to 1) when the counter reaches the input's MSB position. An SRAM stores both the weight and its complement (\weight), and two TGs are used to select between the weight and its complement. The input is processed in bit-serial order from MSB to LSB. If the input bit is zero, the result will be zero. Each DCIM cell has two inputs and two corresponding weights, enabling it to perform multiplication and addition operations. The results are generated through a full adder, which outputs the sum and carry for further processing. The Carry Control block is to generate the additional "1" for carry-in, which is enabled only when input reaches MSB.

C. DCIM Macro and Adders

The proposed DCIM macro is illustrated in Fig. 6, which comprises a two's complement recurrent input computation block, a gated-out SRAM block for ECG input and a shift register. The shift register selects the column of partial sums to send to adder tree, as shown in Fig. 7. In conventional DCIM design, each column of partial sums requires an adder tree for aggregation, leading to hardware inefficiencies. As the objective of ECG classification is to accurately categorize the current heartbeat before the arrival of the next one, it is crucial to maintain a low latency. To trade-off between latency and hardware cost, our proposal involves reusing a single adder tree at the cost of sacrificing clock cycles. A global counter is required to control the overall circuit. This counter keeps track of the bit-length of the input signal, which is set to 5, and

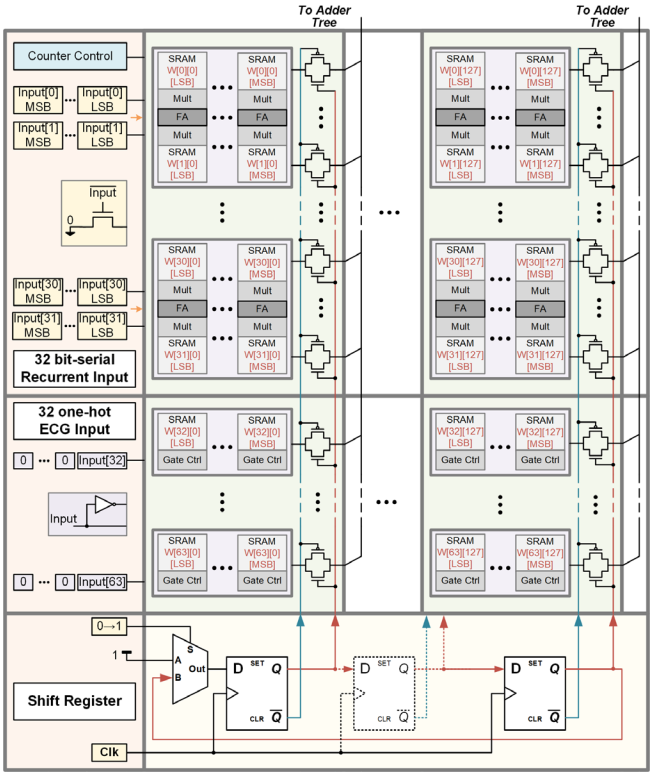


Fig. 6. Proposed DCIM macro architecture featuring 2's complement multiplication for ECG signal processing on LSTM neural network.

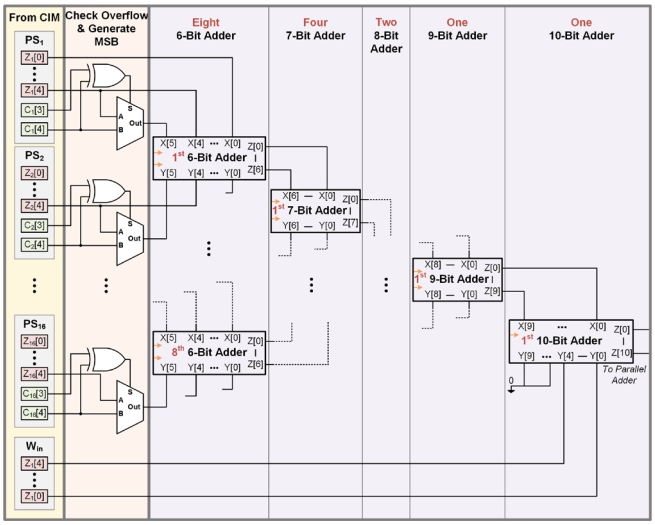


Fig. 7. Adder tree to compute the bitwise summation from a column of partial sums and send to parallel adder.

generates the clock signal transmitted to the shift registers, as shown in Fig. 6. These shift registers are responsible for implementing the reuse of the adder tree.

Figure 7 illustrates an adder tree designed to sum outputs from multiple DCIM cells, featuring a hierarchical structure to manage bit-width expansion across stages. The tree begins with six-bit adders that take initial inputs from CIM outputs. Each subsequent stage increases the bit width to accommodate larger sums. Overflow checks and MSB generation are performed at the input stage to handle carry propagation effectively. The final output from the adder tree is sent to a parallel adder, enabling a cumulative result from all intermediate sums. Since the critical path of the multiplication

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on September 29,2026 at 12:18:01 UTC from IEEE Xplore. Restrictions apply.

TABLE I
BENCHMARK TABLE

	JSSC 2021 [4]	VLSI 2021 [16]	JSSC 2022 [17]	ISCAS 2023 [18]	**This Work**	
Technology	65 nm	65 nm	65 nm	180 nm	28 nm	
Bitcell	20T	6T SRAM	8T	12T	10T	8T
Bitcell Area	6.19 μm^2	NA	3.24 μm^2	NA	0.62 μm^2	0.50 μm^2
Cell Area	10.53 μm^2	0.30 $mm^{2\,**}$	NA	NA	10.53 $\mu m^{2\,*}$	2.48 $\mu m^{2\,*}$
Array Size	128×128	64×64	128×128	9×N	32×640	32×640
Bitcell Density	16 Kb	4 Kb	16 Kb	NA	20 Kb	20 Kb
Input Precision	16-bit in serial	4-bit in serial	5-bit in serial	1-bit	5-bit in serial	1-bit
Output Precision	1-to-16 bit	4-bit	Unsigned 5-bit	1-bit	Signed 5-bit	Signed 5-bit
Weight Precision	1-to-16 bit	4-bit	1	1-bit	Signed 5-bit	Signed 5-bit
Operation Mode	Fully DCIM	Fully DCIM	DCIM + Analog Accumulation	SRAM-based CIM	Fully DCIM	
2's Complement Multiplication	Yes	No	No	No	Yes	
Application	ImageNet	GSCD	MNIST	Convolution	ECG Classification	
Model Topology	AlexNet	LSTM	MLP	CNN	LSTM	
Performance	83.6% Accuracy	1.52x Speedup	96.4% Accuracy	102 µs Latency per Image	96.6% Accuracy	

* Estimated ** Calculated based on information provided in paper

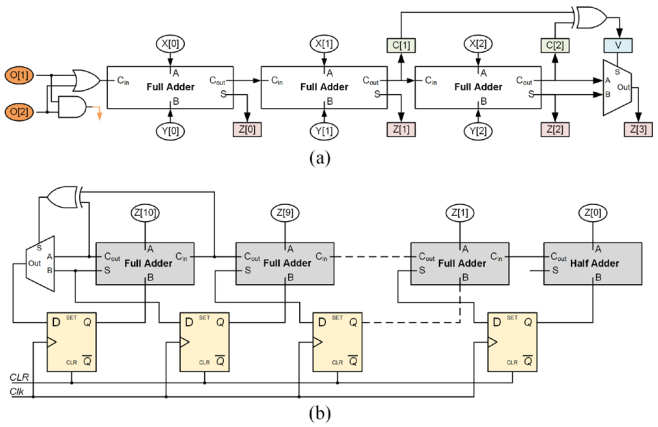


Fig. 8. (a) Two's complement adder in adder tree, using 3-bit as an example. (b) Parallel adder whose Clk is controlled by global counter.

within the DCIM cell involves handling two negative inputs, the multi-bit adder in the adder tree is designed to include a Carry Control block, as shown in Fig. 8(a). This allows the additional carry-in from the two negative inputs to be propagated to subsequent stages. In addition, the schematic diagram of the parallel adder is shown in Fig. 8(b), where the clock (Clk) is controlled by the global counter. After each cycle, the computed LSB is ignored, as only the 6 MSBs of the total 12-bit output are needed. Furthermore, a half-adder is used for the LSB, since there is no carry-out from the shift-and-add operation in the parallel adder.

III. RESULTS AND DISCUSSION

The performance of the proposed framework is evaluated with the MIT-BIH ECG database which includes 48 sets of 30-min-long ECG data [12]. The ECG signal is segmented into full heartbeats based on Christov's peak detection algorithm [14]. Each dataset contains both normal and abnormal ECG features (N, S, V, F, Q) defined in the AAMI standard [15]. For software implementation, the neural network weights are normally stored in 32-bit floating point format (FP32). However, as the goal of this work is to implement this algorithm in resource-constraint portable devices, quantization of network parameters becomes necessary. In addition to the input signal quantization through multi-level one-hot encoding, all weights in the subsequent LSTM layers, FC layer, recurrent outputs in the LSTM layer and interlayer outputs are quantized into signed INT5. Both Post-Training Quantization (PTQ) and Quantization-Aware Training (QAT) techniques are employed to achieve 96.9% accuracy for 5-class classification task.

As the weight matrix multiplication in the LSTM layer is performed using DCIM, the multiplication results are retained in full precision, resulting in minimal error in hardware implementation. The performance of the proposed model is summarized and compared with other recent state-of-arts designs in Table I. The DCIM macro is simulated using 28 nm technology, where two types of bitcells are designed to process ECG signal on LSTM neural network effectively. As shown in Fig. 6, each cell for recurrent input consists of two 5-bit weights and their MAC operations, with every column containing two 10T bitcells and a 14T full adder. The cell area for computing encoded one-hot ECG input includes five 8T bitcells. This design employs fully DCIM operations that support two's complement multiplication, using 10 fewer transistors per bitcell compared to [4].

IV. CONCLUSION

In this paper. this work introduces a DCIM macro optimized for ECG signal processing with an LSTM neural network. By utilizing two bitcell computing mechanisms for recurrent and encoded one-hot inputs, the architecture achieves high accuracy of 96.9% while remaining compact and energy-efficient for real-time, portable ECG monitoring. The DCIM approach balances precision and hardware efficiency by reusing a single adder tree, meeting latency and computational needs without accuracy loss. The DCIM approach exhibits potential in reducing latency and maintaining matrix multiplication precision, enabling seamless software-to-hardware transition with exceptional accuracy. The next step is to implement both DCIM macro and classification framework in CMOS ASIC format to provide ECG anomaly classification in real-time.

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on September 29,2026 at 12:18:01 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] Q. Liu *et al.*, "33.2 A fully integrated analog ReRAM based 78.4 TOPS/W compute-in-memory chip with fully parallel MAC computing," in *IEEE Int. Conf. Solid-State Circuits Conf. (ISSCC)*, 2020, pp. 500-502.

[2] F. Cai *et al*., "A fully integrated reprogrammable memristor-COMS system for efficient multiply-accumulate operations," *Nat. Electron.*, vol. 2, pp. 290-299, 2019.

[3] F. Tu *et al*., "ReDCIM: Reconfigurable digital computing-in-memory processor with unified FP/INT pipeline for cloud AI acceleration," *IEEE J. Solid-State Circuits*, vol. 58, no. 1, pp. 243-255, 2023.

[4] H. Kim, T. Yoo, T. T.-H. Kim, and B. Kim, "Colonnade: A reconfigurable SRAM-based digital bit-serial compute-in-memory macro for processing neural networks," *IEEE J. Solid-State Circuits*, vol. 56, no. 7, pp. 2221-2233, 2021.

[5] H. Kim, J. Mu, C. Yu, T. T.-H. Kim, and B. Kim, "A 1-16b reconfigurable 80kb 7T SRAM-based digital near-memory computing macro for processing neural networks," *IEEE Trans. Circuits Syst. I*, vol. 70, no. 4, pp. 1580–1590, 2023.

[6] Y. Su, H. Kim and B. Kim, "CIM-Spin: A scalable CMOS annealing processor with digital in-memory spin operators and register spins for combinatorial optimization problems," *IEEE J. Solid-State Circuits*, vol. 57, no. 7, pp. 2263-2273, 2022.

[7] J. Hu, W. L. Goh and Y. Gao, "Classification of ECG anomaly with dynamically-biased LSTM for continuous cardiac monitoring," in *IEEE Int. Symp. Circuits Syst. (ISCAS)*, 2023, pp. 1-5.

[8] J. Hu, C. S. Leow, W. L. Goh and Y. Gao, "Energy efficient software-hardware co-design of quantized recurrent convolutional neural network for continuous cardiac monitoring," in *Proc. IEEE Int. Conf. Artif. Intell. Circuits Syst. (AICAS)*, 2023, pp. 1-5.

[9] S. S. Virani *et al.*, "Heart disease and stroke statistics—2021 update: A report from the American heart association," *Circ.*, vol. 143, no. 8, 2021.

[10] B. Parhami, *Computer Arithmetic: Algorithms and Hardware Designs*. Oxford University Press, 2010.

[11] J. Hu, W. L. Goh, and Y. Gao, "LSTM-based ECG signal classification with multi-level one-hot encoding for wearable applications," in *IEEE Biomed. Circuits Syst. Conf. (BioCAS)*, 2024, pp. 1-5.

[12] G. Moody and R. Mark, "The impact of the MIT-BIH arrhythmia database," *IEEE Eng. Med. Biol. Mag.*, vol. 20, no. 3, pp. 45–50, 2001.

[13] Hochreiter, S., and J. Schmidhuber. "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp.1735-1780, 1997.

[14] I. I. Christov, "Real time electrocardiogram QRS detection using combined adaptive threshold," *Biome. Eng.*, vol. 3, no. 1, pp. 1–9, 2004.

[15] AAMI/ANSI, Testing and Reporting Performance Results of Cardiac Rhythm and ST-segment Measurement Algorithms. The Association, 1999.

[16] R. Guo *et al*., "A 6.54-to-26.03 TOPS/W computing-in-memory RNN processor using input similarity optimization and attention-based context-breaking with output speculation," in *IEEE Symp. VLSI Circuits (VLSI)*, 2021, pp. 1-2.

[17] C. Yu, T. Yoo, K. T. C. Chai, T. T. -H. Kim and B. Kim, "A 65-nm 8T SRAM compute-in-memory macro with column ADCs for processing neural networks," *IEEE J. Solid-State Circuits*, vol. 57, no. 11, pp. 3466-3476, Nov. 2022.

[18] G. Domenech-Asensi, R. Ruiz-Merino, J. Zapata-Perez and J. A. Díaz-Madrid, "A 12T SRAM in-memory computing differential current architecture for CNN implementations," in *IEEE Int. Symp. Circuits Syst. (ISCAS)*, 2023, pp. 1-5.

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on September 29,2026 at 12:18:01 UTC from IEEE Xplore. Restrictions apply.